

AHRI 1380

Updating to "-2026" from "-2019"

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Proposed Goals for AHRI 1380-2026

Define a standard that enables HVAC systems to participate in demand flexibility, in order to:

- Provide customers the comfort they desire, at the **lowest energy cost**
- Coordinate with the electric grid to **reduce peak load events**, lowering infrastructure costs
- **Shift energy usage** to minimize both environmental impact (GHG) and cost

Enable demand flexibility control by any or all of:

- Customer
- Utility
- Aggregator (including the HVAC manufacturer acting as aggregator)

AHRI 1380-2019 Protocols

OpenADR 2.0b

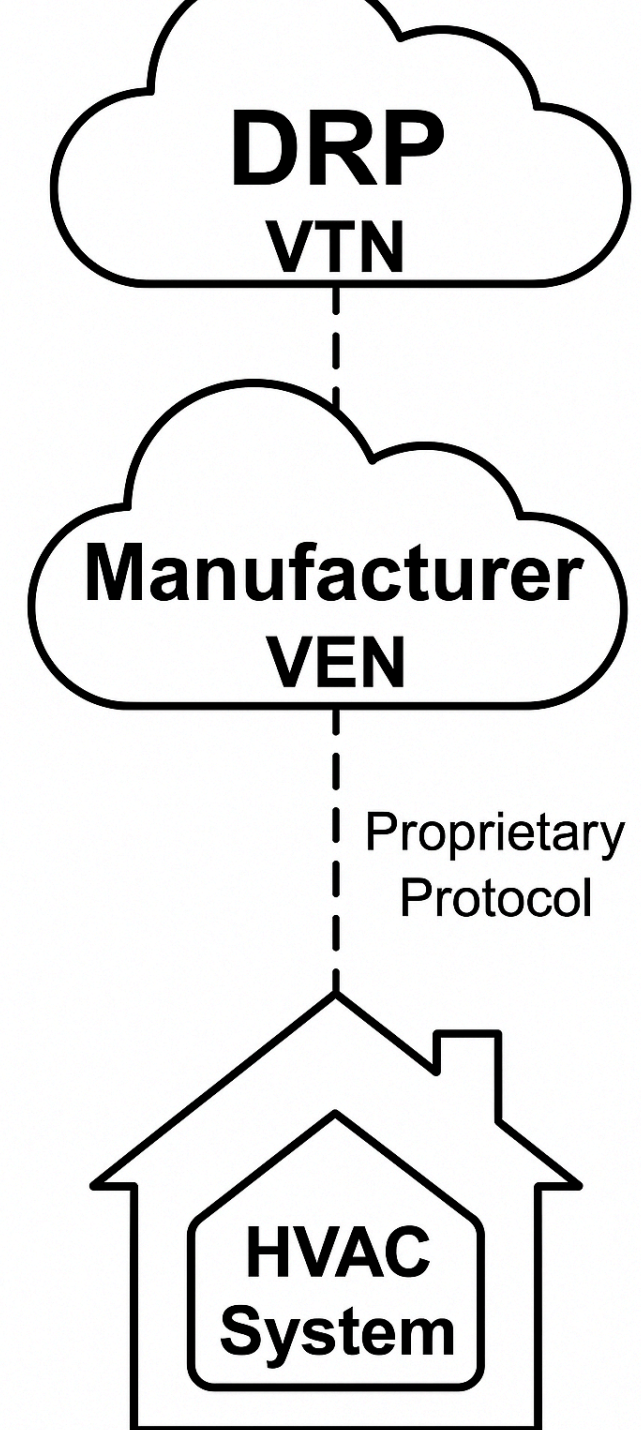
- HVAC system connects via Wi-Fi and/or Ethernet
- VEN (client) typically resides in **manufacturer's cloud**
- Connection between HVAC system and cloud uses proprietary protocol

| *Are there ANY native-VEN HVAC systems?*

CTA-2045-A

- A port enabling an aftermarket UCM to control the HVAC system
- UCM's connection to DRP uses unspecified protocol

| *Are there ANY HVAC systems that support CTA-2045?*



Educated Guesses & Assumptions

Virtually all AHRI 1380-2019 compliant HVAC systems do so via:

- OpenADR 2.0b VEN residing in the **manufacturer's cloud**
- Wi-Fi interface on HVAC system
- **No** HVAC systems support OpenADR native VENs
- **No** HVAC systems support CTA-2045 ports

Industry Trends

- **Dynamic pricing** is superior to traditional DR events — CPUC & CEC transitioning
- Flexible-demand appliances per household are **growing** — need coordinated management
- Grid coordination transitioning from **device** to **site**
 - Dynamic power import/export limits per site (NEC-705.13 & UL-3141)
 - Site management will occur **locally**, not in the cloud
- **OpenADR 3.1** supports everything 2.0b does, plus:
 - Both cloud-to-cloud AND cloud-to-device communication
 - Simpler & easier to develop and deploy
- **Matter** adopted for smart-appliance interoperability (800+ companies)
 - Does not provide grid-to-customer coordination alone
 - Combined with OpenADR 3.1: local HEMS controls via Matter

Functional Control and Demand Response

- Device behavior is altered via **functional-control commands**
- Optimal translation requires **local customer context**
- The translation location must be **customer's choice**: device, local HEMS, or ASP
- Translation **must not** be locked in the manufacturer's cloud

Local HEMS is the best location to translate price & DR events to functional-control commands

Ensured vs. Enabled

Ensured

The HVAC system supports:

- An open/standard protocol for flexible demand control
- Server URL is customer-configurable
- Network interface(s) for LAN/Internet

Fully capable of participating in demand flexibility with no additional cost or effort

Enabled

The HVAC system supports:

- A connector/port for another device
- That device provides flexible demand control by unspecified means

Requires additional cost and effort to participate

Assessing AHRI 1380-2019

Compliance Type	Mechanism
Ensured	Wi-Fi + OpenADR 2.0b VEN in manufacturer cloud
Enabled	CTA-2045-A

Experience with CTA-2045 for water heaters has clearly demonstrated that the cost and installation of a UCM is too large, resulting in very poor connection rates. CTA-2045 is obsolete and should be EOL'd.

Implications from Trends

- **Begin transition** from OpenADR 2 to OpenADR 3.1
- **Define native HVAC system control** to enable:
 - Coordinated management of all home appliances
 - Allocation of site capacity limits to specific appliances
- **Add Matter support** to native HVAC systems for:
 - Functional control by Matter-based HEMS
 - Customer integration into "whole home" management apps
- **Drop CTA-2045 support:**
 - Few/no HVAC systems support it
 - Has failed for water heater DR with plenty of time to succeed

AHRI 1380-2026 Proposal: Networking

HVAC systems must support (or be capable of supporting):

- Wi-Fi
- Cellular modem
- Ethernet

Implementation:

- HVAC system may directly integrate any or all interfaces
- HVAC system **must** provide a **USB-C port** for network interface dongles
- Customer (or DRP) adds non-integrated interfaces as needed

6.2.2 *Requirements for connecting to OpenADR 2.0.* DR-ready HVAC System using the OpenADR 2.0 option shall include one or more of the following interfaces. Location of the interface can reside within any of the required system architecture.

6.2.2.1 IEEE/ISO/IEC 8802-2 (Ethernet).

6.2.2.2 IEEE 802.11 (Wi-Fi) with the option of access to a Cloud Server.

USB-C Network Dongles



USB WiFi Adapter, Plug-n-Play, Built-in high-gain Antenna.USB Adapter for PC Desktop/Laptop, Wireless Network Adapter Supports Windows 11/10/8/7

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Amazon Basics Aluminum USB-C to RJ45 Gigabit Ethernet Adapter, Fast Network, Grey, 2.07 x 0.81 x 0.6 inches

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Inexpensive Ethernet dongle

AHRI 1380-2026 Proposal: Protocols

Two distinct compliance types — a system **must** support one, **may** support both:

Native

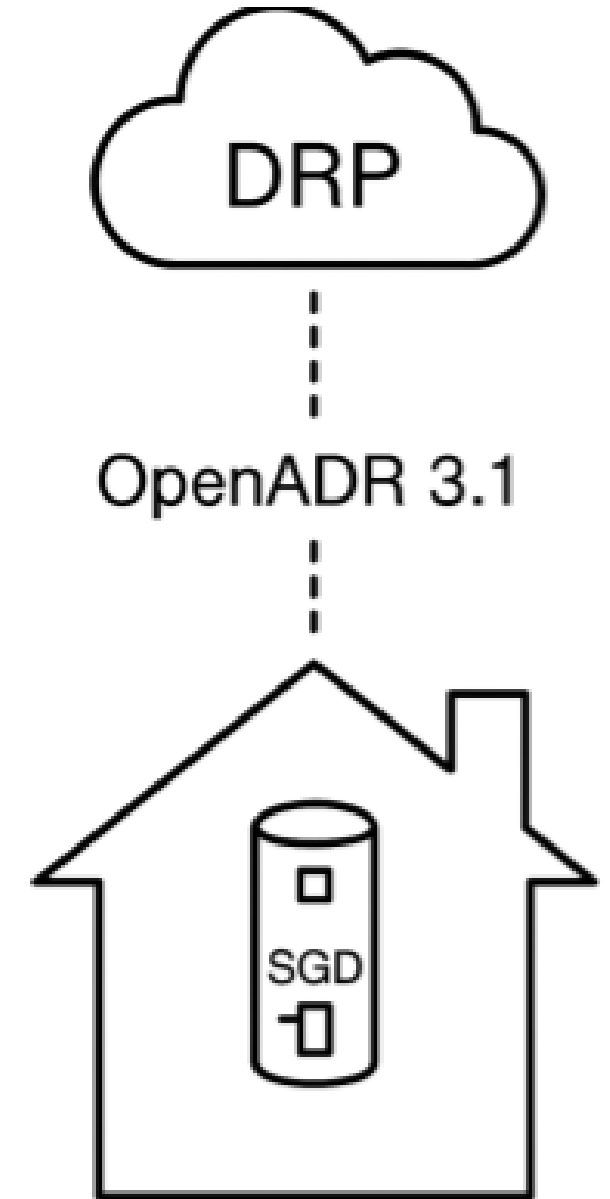
Protocol(s) provided directly & locally by the HVAC system:

- **OpenADR 3.1 VEN** — Mandatory (MUST), VTN URL customer-configurable
- **Matter 1.4+** — Optional (SHOULD): DEM, DEM Mode, Thermostat clusters

Cloud

Protocol(s) provided via the manufacturer's cloud:

- **OpenADR** — Mandatory (MUST)
 - 3.1 VEN: Preferred (SHOULD)
 - 2.0b VEN: Allowed (MAY) during transition



Compliance/Certification Options

Option	Protocol	Networking
2026-Cloud + 2019-Networking	OA 2.0b or 3.1 in cloud	Wi-Fi and/or Ethernet
2026-Cloud + 2026-Networking	OA 2.0b or 3.1 in cloud + optional HCA	Integrated + USB-C
2026-Native + 2026-Networking	OA 3.1 integrated + optional Matter	Integrated + USB-C

A compliant HVAC system must specify which option(s) it supports.

Enables existing products to comply while providing a clear transition path.

Where the Puck Is Going: Grid Coordination

- "Macro" coordination will transition from DR events to **highly dynamic pricing**
- "Micro" coordination will support **dynamic capacity management** per site
 - "Current price is P \$/kWh, your maximum site import is N Amps"
- HEMS will **autonomously** control loads based on price, capacity limits, customer preference, local generation, and storage

Native control of HVAC power & energy is the only way to support these use cases.

HVAC manufacturers should focus on and prioritize native protocol support.

OpenADR 3.1 supports BOTH cloud-to-cloud AND cloud-to-native use cases.

Where the Puck Is Going: HVAC Systems

From	To
Natural gas	Electricity (heat pump)
Fixed speed	Variable speed (inverters)
One zone	Multiple zones (or mini-splits)
DR client in manufacturer cloud	DR client in HVAC system
Proprietary control protocol	Open & standard protocol
Controller in thermostat	Controller in air handler
Thermostat UI	Tablet & phone apps
Central thermostat sensor	Distributed IoT sensors



CEC and Demand Flexibility

The CEC has begun a series of dockets to define **Flexible Demand Appliance Standards (FDAS)**:

- **Docket 24-FDAS-03** for Low-Voltage Thermostats (September 2024)
- **Docket 24-FDAS-04** for Electric Vehicle Supply Equipment (October 2024)

It behooves AHRI 1380 to support potential regulatory mandates and future rebate/incentive requirements.

The 1380-2026 proposals provide a **smooth transition path** from today to tomorrow, enabling HVAC systems to "skate to where the puck is going."

Exit Protocols & MITO

Exit Protocol ("Snapback")

- Simultaneous startup of many HVAC systems after DR event stresses the grid
- Solutions: randomized duration offsets, staggered cancellation events
- Likely a server implementation issue, not a protocol issue

Maximum Indoor Temperature Offset (MITO)

Most useful HVAC controls for DR:

- **Mode:** Heat pump only, vacation, off
- **Setpoint:** Raise (cooling) or lower (heating) to attenuate power
- **Power adjustment:** Variable-speed systems support throttle control
- **Zone:** Mode and setpoint per zone

Variable power adjustment via Matter DEM and DEM-mode clusters provides the most granular control. Local control is crucial for HEMS.